

Vittorio Maria Ruffo

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Education

Frankfurt School of Finance & Management

Frankfurt, Germany

PHD IN FINANCE AND MASTER IN BUSINESS RESEARCH AND ANALYTICS

September 2022 - present

- Main research interests: empirical asset pricing, applications of AI, volatility, decentralized finance
- Advisor: Prof. Grigory Vilkov

Collegio Carlo Alberto and University of Turin

Turin, Italy

SECOND LEVEL SPECIALIZING MASTER IN FINANCE, INSURANCE AND RISK MANAGEMENT

September 2021 - July 2022

- Merit based full tuition waiver

University of Verona

Verona, Italy

MASTER IN BANKING AND FINANCE - QUANTITATIVE FINANCE TRACK

September 2019 - September 2021

- Final Grade: 110/110 with honours
- Supervisor: Prof. Alessandro Buccioli
- Thesis in Econometrics “*Personality and investment choices: an analysis on Australian panel data*”

University of Verona

Verona, Italy

BACHELOR IN ECONOMICS AND BUSINESS

September 2016 - September 2019

- Final Grade: 110/110 with honours
- Supervisor: Prof. Elena Giarretta
- Thesis in Business Management “*Tourism between sustainability and innovation: the “Glamping” phenomenon*”

Working Papers

Market Efficiency in Prediction Markets: A Comparison with Derivatives, with M. Fabi, R. Marfè, and L. Schoenleber [link to paper]

Abstract: We study pricing efficiency in decentralized prediction markets by comparing Polymarket prices with option-implied prices derived from derivatives. We analyze nearly 5,000 Polymarket contracts written on Bitcoin (BTC) and Ethereum (ETH), two assets for which prediction-market contracts and liquid derivatives markets coexist. We find that Polymarket prices closely track option-implied prices, but systematic deviations remain: they are concentrated in tail and barrier contracts, vary with sentiment, volatility, demand, and market frictions, and imply more probability mass on extreme price movements. To guide our empirical investigation, we introduce a parsimonious limits-to-arbitrage model that formalizes equilibrium behavior in decentralized prediction markets.

Factor Dispersions, with D. Gerchik, L. Schoenleber, and G. Vilkov [link to paper]

Abstract: Dispersion strategies capture the difference in variance dynamics between a basket and its components. Even though smart-beta indices intend to load heavily on a particular factor, factor dispersions based on such baskets are exposed to risks of other factors and idiosyncratic variances. Analyzing factor dispersions through a linear factor model and equicorrelation representations, we recover driving forces behind dispersion dynamics and work out an attribution of a dispersion risk premium. As a balanced combination of systematic and idiosyncratic variance components, dispersion and its risk premium provide signals about future changes in systematic and alpha-based investment opportunities.

- Winner of the CBOE - The Options Institute - S&P Dow Jones Indices Dispersion Research Grant

Work in Progress

Firm-level News Networks, with C. Dim and G. Vilkov

Presentations (* scheduled)

June 2026. (*) 8th Future of Financial Information Conference, Frankfurt, Germany.

May 2026. 3rd Structured Retail Products and Derivatives Conference, Hagen, Germany.

May 2026. Designing DeFi Conference, New York, USA.

May 2026. 2026 conference Tech 4 Finance 3: AI and Blockchain, Paris, France.

April 2026. XXVII Workshop on Quantitative Finance, University di Bergamo, Bergamo, Italy.

January 2026. Decentralized Finance & Crypto Workshop, Scuola Normale Superiore, Pisa, Italy.

December 2025. 2025 Global AI Finance Research Conference, Hong Kong, China.

July 2025. Second Liverpool Workshop in Option Markets, Liverpool, UK.

November 2024. 2024 FMA Conference on Derivatives and Volatility, Chicago, USA.

May 2024. Bonn-Frankfurt-Mannheim PhD Conference 2024, Bonn, Germany.

Teaching and Work Experience

- 2026- Lecturer, Corporate Finance (Bachelor), Frankfurt School
- 2025 TA in Portfolio Optimization in Continuous Time (Master, Prof. Francesco Sangiorgi), Frankfurt School
- 2025- Lecturer, Introduction to Economics and Finance (High school), Associazione Lagrange
- 2025- Lecturer, Python Bootcamp (Master), Teaching rating (4.1/5), Frankfurt School
- 2024- TA in Financial Products & Modelling (Master, Prof. Grigory Vilkov), Frankfurt School
- 2024- TA in Corporate Finance (Bachelor, Prof. Larissa Schaefer and Prof. Yigitcan Karabulut), Frankfurt School
- 2024 RA for Quantitative Asset Management, Prof. Francesco Sangiorgi, Frankfurt School
- 2023 TA in Portfolio Management (Master, Prof. Paula Cocoma), Frankfurt School
- 2022 Teacher, Data Analysis Laboratory with R, University of Verona
- 2021 TA in Financial Econometrics (Master, Prof. Alessandro Buccioli), University of Verona
- 2021 TA in Statistics (Bachelor, Prof. Marco Minozzo), University of Verona
- 2020-2021 TA in Business Management (Bachelor, Prof. Elena Giaretta and Prof. Angelo Bonfanti), University of Verona

Scholarships

- 2022 Doctoral Scholarship Frankfurt School of Finance & Management
- 2021 Merit based full tuition waiver Collegio Carlo Alberto

Computer Skills

Python, Matlab, R, Stata, SQL, Java, Latex, Adobe Lightroom and Photoshop, Phase One Capture One

Additional Information

Languages: English (Fluent), Italian (Native speaker)

Interests: Photography, Hiking, Cooking